

# The Rate Hinge

Change in an accumulated quantity per unit time

IB Physics – student worksheet

## Shared architecture

$$\text{average rate} = \frac{\Delta X}{\Delta t}$$

$$\text{instantaneous rate} = \frac{dX}{dt}$$

The symbol  $X$  represents an accumulated quantity: displacement, charge, transferred energy, or number of undecayed nuclei. The physical meaning of the rate comes from the numerator and its sign.

**For every case:** identify  $X$ , state the rate quantity, calculate its value and units, and decide whether an average or instantaneous rate is being described.

### 1. Motion: velocity as a rate

A trolley moves along a straight track. Its displacement changes from  $x = 1.4\text{ m}$  at  $t = 2.0\text{ s}$  to  $x = 8.6\text{ m}$  at  $t = 5.0\text{ s}$ .

$$\bar{v} = \frac{\Delta x}{\Delta t}$$

- Identify the accumulated quantity  $X$ .
- Calculate the average velocity over the interval.
- Explain why displacement, rather than distance travelled, appears in this definition.
- On a displacement–time graph, what geometrical quantity gives average velocity?
- What geometrical quantity gives instantaneous velocity at one time?
- The trolley reverses direction. Explain how velocity can be negative even though time continues to increase.

#### Completion statement

Velocity is the rate of change of \_\_\_\_\_ with respect to \_\_\_\_\_.

### 2. Charge flow: electric current

A sensor records 18 C of charge crossing a wire section in 6.0 s.

$$I = \frac{\Delta Q}{\Delta t}$$

- Identify the accumulated quantity  $X$ .
- Calculate the average current.
- Express the ampere in base rate form using coulombs and seconds.
- A graph of transferred charge  $Q$  against time curves upward. What does this imply about the current?
- A steady current of 2.5 A flows for 4.0 min. Calculate the charge transferred.
- Explain why current describes flow through a boundary rather than the amount of charge stored at one location.

#### Completion statement

Current is large when charge crosses the chosen section \_\_\_\_\_.

### 3. Energy and nuclei: power and activity

A heater transfers 72 kJ of energy in 3.0 min.

$$P = \frac{\Delta E}{\Delta t}$$

A radioactive sample undergoes 5400 decays in 30 s.

$$A = -\frac{\Delta N}{\Delta t}$$

- Calculate the average power of the heater.
- State the joule-per-second unit name.
- Calculate the sample activity in becquerels.
- Explain why a minus sign may appear in  $A = -dN/dt$ .
- Distinguish activity from the number of undecayed nuclei remaining.
- A source has constant activity over a short measuring interval. Sketch or describe the corresponding graph of cumulative decays against time.

#### Completion statement

Power measures energy transfer rate; activity measures \_\_\_\_\_.

## Synthesis: the denominator is shared; the numerator carries the physics

| Context           | Accumulated quantity $X$ | Rate | Unit | Average or instantane-<br>ous? | Meaning of sign or direction |
|-------------------|--------------------------|------|------|--------------------------------|------------------------------|
| Motion            |                          |      |      |                                |                              |
| Charge flow       |                          |      |      |                                |                              |
| Energy transfer   |                          |      |      |                                |                              |
| Radioactive decay |                          |      |      |                                |                              |

### Final questions

- What is genuinely shared by  $v = dx/dt$ ,  $I = dQ/dt$ ,  $P = dE/dt$ , and  $A = -dN/dt$ ?
- Why does quoting only “per second” fail to identify a physical rate?
- Complete: *The gradient of an accumulated-quantity graph represents* \_\_\_\_\_.

# The Rate Hinge

Teacher guide and worked answers

**Purpose.** Students recognise rates as derivatives or finite-interval ratios while preserving the distinct physical meanings of displacement, charge, energy and nuclear population.  
**Core distinction.** The shared denominator does not make the rates interchangeable. The numerator, boundary, sign convention and measurement interval determine the physical claim.

### 1. Motion – answers

1.  $X = x$ , displacement.
2.
$$\bar{v} = \frac{8.6 - 1.4}{5.0 - 2.0} = \frac{7.2}{3.0} = 2.4 \text{ m s}^{-1}.$$
3. Displacement is signed and records change of position. Distance records total path length and gives average speed instead.
4. The gradient of the secant joining the two points.
5. The gradient of the tangent at that time.
6. A negative velocity means displacement is decreasing under the chosen positive direction. It does not imply negative elapsed time.

#### Teaching emphasis

Keep average and instantaneous quantities separate. A finite interval gives  $\Delta x/\Delta t$ ; a tangent gives  $dx/dt$ .

### 2. Current – answers

1.  $X = Q$ , charge transferred across the selected section.
2.
$$I = \frac{18}{6.0} = 3.0 \text{ A}.$$
3.  $1 \text{ A} = 1 \text{ C s}^{-1}$ .
4. The gradient  $dQ/dt$  is increasing, so the current is increasing.
5.  $4.0 \text{ min} = 240 \text{ s}$ , hence

$$Q = It = (2.5)(240) = 600 \text{ C}.$$

6. Current counts charge crossing a specified boundary per unit time. Charge stored in a region is an inventory, not a flow rate.

#### Teaching emphasis

The boundary must be stated. The same circuit can contain stored charge while supporting zero current, or a steady current with little net local charge accumulation.

### 3. Power and activity – answers

Convert  $72 \text{ kJ} = 7.2 \times 10^4 \text{ J}$  and  $3.0 \text{ min} = 180 \text{ s}$ .

1.
$$P = \frac{7.2 \times 10^4}{180} = 4.0 \times 10^2 \text{ W}.$$
2.  $1 \text{ W} = 1 \text{ J s}^{-1}$ .
3.
$$A = \frac{5400}{30} = 180 \text{ s}^{-1} = 180 \text{ Bq}.$$
4.  $N$  decreases, so  $dN/dt < 0$ . The minus sign makes activity a positive decay rate.
5.  $N$  is an inventory;  $A$  is the rate at which that inventory decreases.
6. A straight line with positive gradient equal to the activity, for cumulative decays plotted against time.

### Completed synthesis

| Context           | Accumulated quantity   | Rate                  | Unit                         | Form used here                | Sign or direction                                                  |
|-------------------|------------------------|-----------------------|------------------------------|-------------------------------|--------------------------------------------------------------------|
| Motion            | displacement $x$       | velocity $v = dx/dt$  | $\text{m s}^{-1}$            | finite interval and tangent   | Sign gives direction along the chosen axis.                        |
| Charge flow       | transferred charge $Q$ | current $I = dQ/dt$   | $\text{A} = \text{C s}^{-1}$ | finite interval               | Sign depends on chosen current direction and charge convention.    |
| Energy transfer   | transferred energy $E$ | power $P = dE/dt$     | $\text{W} = \text{J s}^{-1}$ | finite interval               | Sign can indicate energy entering or leaving a chosen system.      |
| Radioactive decay | undecayed nuclei $N$   | activity $A = -dN/dt$ | $\text{Bq} = \text{s}^{-1}$  | finite interval approximation | Minus sign converts a decreasing inventory into positive activity. |

### Answers to final questions

1. Each is the rate of change of an accumulated quantity with respect to time. The numerator gives the physical interpretation.
2. “Per second” supplies only the denominator. One must state what changes per second and whether it is signed, directional or counted across a boundary.
3. The gradient represents the corresponding instantaneous rate; over a finite chord it represents an average rate.

**Classification.** This is primarily a *bookkeeping hinge*: the mathematical operation organises quantities and units. It does not by itself explain the mechanism producing the motion, current, transfer or decay.